>> Intelligent Systems – WC 7 <<

Machine Learning 2 and extra preparation for 2nd partial exam

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Try to make exercise before you come to WC. By this means you will be aware of your own knowledge about the topics. Good luck!

Also there is a good chance that there will not be enough time to go through all the exercises during the working group meetings. Make sure you know what questions to ask.

Machine Learning 2

Exercise 1: Design Matrixes

Consider the following training set of positive examples {(1,1),(1,-1),(-2,2),(-2,-2)} and negative examples {(3,3),(3,-3),(-4,4),(-4,-4)}. Draw the feature space and try to fit a linear separator between the classes.

What can you do to rectify the problem? Provide a simple design matrix with two combined features instead of the original ones for which there is a simple separator, and give an example for such a linear separator.

The solution is simple and does not need to be formalised. If you choose two features $x' = x^2$ and $y' = y^2$, you get a new positive training set {(1,1),(4,4)} and a negative one. {(9,9),(16,16)}. Any line between those two sets is a linear separator, e.g. $f(x') = 6$.

The easiest is to simply draw the coordinate systems for the original feature set, and the design matrix.

Exercise 2: Linear Regression

Here we are going to practise fitting a linear function to a number of training sets. Suppose we have three pairs of values (1,1), (2,3) and (3,2) for two features. As discussed in the lecture, according to the ML recipe, the results depend on the chosen model class. Let us calculate the optimal function according to a number of different classes.

2a) Let us start with the most simple model class to fit our training examples, a constant line. This means that we are looking at a function $f(x) = b$, which is a horizontal line in the coordinate system. What is the result of learning the optimal function f(x) from the given data? Hint: we are looking for a value for b.

Solve this problem by using the following steps:

1. Calculate the value of the (squared) error function for each of the three points
2. Calculate the error function based on the points for the three points
3. Search for the minimal value for this function, by
    a. Calculating the derivation of the function w.r.t. b
    b. Setting this derivation to 0, and solve the equation for b.

The value you find for b gives you the function, i.e. the optimal horizontal line fitting your data.

1) The value of the squared error function for point (1,1) is $(b-1)^2$, as the predicted value for the function f(x) is b, and the example is 1. We square to get a positive value. Similarly for (2,3), we get $(b-3)^2$ as the expected value is b and not 3, and finally $(b-2)^2$ for (3,2). You see that the value of x does not play a role here, as we are looking for a function where the slope is 0, so we always would add 0*1, 0*2 and 0*3 for the three points.
2) The overall loss is then just the sum over the three points, $err(p) = (b-1)^2 + (b-3)^2 + (b-2)^2$ This is calculated to be $err(p)=3b^2 -12b + 14$.
3) To find the minimum function f(x) =b we have to
   a) calculate the derivation of the error function, i.e. $err'(p) = 6b – 12$, and
   b) set it to 0. This means 0=6b-12, which is the same as b=2..

This means that the function f(x) = 2 is the optimal fitting function for our training set.

2b) A slightly more complex example is a line with a given, constant slope. Here we choose e.g. s=1. This means that we are looking at a function f(x) = 1*x + b, which is a line in the coordinate system with a slope of 45 degree. What is the result of learning the optimal function f(x) from the given data? Hint: we are still only looking for a value for b.

Follow the same method as described for 2a.

1) The value of the squared error function for point (1,1) is $(b+1-1)^2$, as the predicted value for the function f(x) is b+x, and the example is (1,1). We square to get a positive value. Similarly for (2,3), we get $(b+2-3)^2$ as the expected value is b+2 and not 3, and finally $(b+3-2)^2$ for (3,2). You see that the value of x is simply added here, as we are looking for a function where the slope is 1*x, so we add 1*1, 1*2 and 1*3 for the three points.
2) The overall loss is then just the sum over the three points, $err(p) = b^2 + (b-1)^2 + (b+1)^2$ This is calculated to be $err(p)=3b^2$.
3) To find the minimum function f(x) =b we have to
   a) calculate the derivation of the error function, i.e. $err'(p) = 6b$, and
   b) set it to 0. This means b=0.

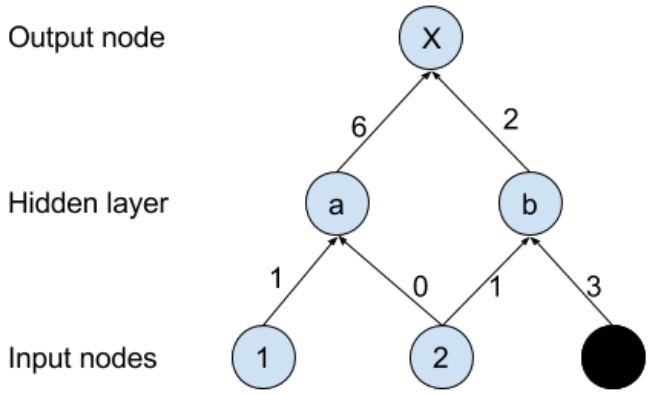
This means that the function f(x) = x + 0 = x is the optimal fitting function for our training set.

Exercise 2c (Not tested in the exam) The most interesting case for this simple example is, of course, when we also want to find the optimal slope s, i.e. a full linear function f(x)=s*x+b. The procedure is the same, only that the error function now also depends on s. This means you will have to calculate two partial derivatives, one for s and one for b. Setting both of them to be 0 will give you the optimal values for s and b. Try it out, but working with partial derivatives will not be tested in the exam.

<u>Exercise 3: Neural Networks</u>

Here, we see a basic multilayer feedforward neural network, with weights and values on the input nodes. It has no activation function (all nodes are *linear*). The black node is a *bias node*. What is the value on the output node (marked X)?

I



This is a very simple exercise: the value of a node y is by definition calculated by the values of its children xi multiplied by their weights wi, so with two nodes a and b we have y= w1*x1 + w2*x2. This means the value for node a is calculated as 1 * 1 + 0 * 2 = 1, for b as 1* 2 + 3*1 = 5. Remember that the bias node always has the value 1. We do the same for X, which is calculated as 6 * 1 + 5* 2 = 16. That's all.

Recap and Exam Preparation

Exercise 4: Fuzzy Logic

Given the following (standard) semantics for Fuzzy Logic for the following operators:

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A),f(B))
- f(A & B) = min(f(A),f(B))
- f(A -> B) = min(1;1 - f(A) + f(B))

What would be the fuzzy interpretation of the formula ((P -> ¬ Q) -> (Q v ¬P)) given fuzzy values f(P)=0.9 and f(Q)=0.2 for P and Q.

Solution:

f(¬Q) = 1 - f(Q) = 1 - 0.2 = 0.8

f(P -> ¬Q) = min(1; 1 - f(P) + f(¬Q)) = min(1; 1 - 0.9 + 0.8) = min(1; 0.9) = 0.9

f(¬P) = 1 - f(P) = 1 - 0.9 = 0.1

f(Q v ¬P) = max(f(Q); f(¬P)) = max(0.2, 0.1) = 0.2

f((P -> ¬ Q) -> (Q v ¬P)) = min(1; 1 - f(P -> ¬Q) + f(Q v ¬P)) = min(1; 1 - 0.9 + 0.2) = min(1; 0.3) = 0.3

Fill in the values for the following subformulas.

f(P -> ¬ Q)= 0.9

f(Q v ¬P)= 0.2

f((P -> ¬ Q) -> (Q v ¬P)) = 0.3

Exercise 5: Fuzzy Relations

Consider the following relations, where WGS-5 and WGS-6 stands for Working Group Session 5 and Working Group Session 6, respectively.

Understanding:

	WGS5	WGS6
Joana	0.3	0.7
Andreea	0.6	0.4
Francesca	0.8	0.2

Contains;

	Fuzzy Logic	Probability
WGS-5	0.7	0.3
WGS-6	0.4	0.6

Fill in the following table for the composition of the two relations:

	Fuzzy Logic	Probability
Joana		
Andreea		
Francesca		

Solution:
You need to apply the following formula for each of the combinations:

$$f_{R\square S}(\langle x,z\rangle) = \max_{y\in Y} \min(f_R(\langle x,y\rangle), f_S(\langle y,z\rangle))$$

So, $f_{\text{understanding o contains}}$ (<Joana, fuzzy logic>)

$= \max_{\text{WGS5, WGS6}} \min(f_{\text{understanding}}(\text{Joana, WGS5}), f_{\text{contains}}(\text{WGS5, fuzzy logic})),$
$\min(f_{\text{understanding}}(\text{Joana, WGS6}), f_{\text{contains}}(\text{WGS6, fuzzy logic}))$
$= \max_{\text{WGS5, WGS6}} \min(0.3, 0.7), \min(0.7, 0.4) = \max_{\text{WGS5, WGS6}} 0.3, 0.4 = 0.4$

And the same for the other fields:

Joana, probability = max(min(0.3, 0.3),min(0.7, 0.6)) = 0.6 -> this is informal
Andreaa, fuzzy logic = max(min(0.6, 0.7), min(0.4, 0.4)) = 0.6
Andreaa, probability = max(min(0.6, 0.3), min(0.4, 0.6)) = 0.4

Francesca, fuzzy logic = max(min(0.8, 0.7) ,min(0.2, 0.4)) = 0.7
Francesca, probability = max(min(0.8, 0.3), min(0.2, 0.6)) = 0.3

	Fuzzy Logic	Probability
Joana	0.4	0.6
Andreea	0.6	0.4
Francesca	0.7	0.3

Exercise 6: Calculating with Joint Probability distributions

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

Given the full joint distribution as above, calculate the following:

7a) P(cavity, toothache | catch) = 0 .108 / 0.108+0.016+0.72+0.144
7b) P(-toothache | cavity) = 0.072+0.008 / 0.108+0.012+0.072+0.008
7c) P(-cavity, toothache, catch) = 0.016

Exercise 7: Bayesian classifier

There is a clear correlation between the weather, whether the NS operates smoothly and my level of frustration. The following table summarises some observations on five different days: Note that the table in the question is a bit ambiguous, whether Trains cancelled NO means that there are trains or not. In the answer it means that there are no trains, so the table below is adapted.

	Cold	Trains	Snow	Wind	Classification: Frustrated
D1	Yes	No	No	Yes	No
D2	Yes	Yes	Yes	No	Yes
D3	No	Yes	No	Yes	Yes
D4	No	No	No	No	No
D5	Yes	Yes	No	Yes	No

Apply Naive Bayesian learning with a naïve smoothing of 0.01 on this data to predict whether I will be frustrated on a day where: it is not cold and no train is cancelled although it is windy and it snows.

Solution:
We want to calculate argmax_{F,-F}(P(F|-C,-T,W,S),P(-F|-C,-T,W,S))

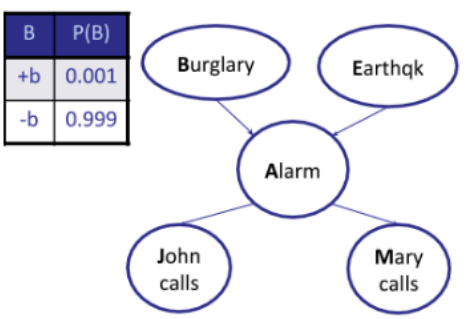
so
P(F|-C,-T,W,S) = P(F) * P(-C|F) * P(-T|F) * P(S|F) * P(W|F)
P(-F|-C,-T,W,S)= P(-F) * P(-C|-F) * P(-T|-F) * P(S|-F) * P(W|-F)

These can be estimated from the table:
P(F) = 2/5
P(-F) = 3/5
P(-C|F) = 1/2
P(-T|F) = 0.01
P(S|F) = 1⁄2
P(W|F) = 1⁄2
P(-C|-F) = 1/3
P(-T|-F) = 2/3
P(S|-F) = 0.01
P(W|-F) = 2/3

This implies:
P(F|-C,-T,W,S) = 2/5 * 1/2 * 0.01 * 1/2 * 1/2 = 1/2000 = 0.0005
P(-F|-C,-T,W,S)= 3/5 * 1/3 * 2/3 * 0.01 * 2/3 = 4/4500 = 1/1125 = 0.00089

P(-F)>P(F) (0.00089 > 0.0005), so the prediction is Not Frustration.

<u>Exercise 8: Bayesian Networks</u>
Given the following Bayesian network:



B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

What is the probability that both John and Mary call, given that the alarm rings?

The solution is 0,63. This is because P(J,M|A) = P(J|M,A) * P(M|A) because of the product rule. As J and M are independent P(J|M,A)=P(J|A)* P(M|A). So, P(J,M|A)=0.9*0.7=0.63

Advanced: What is the probability that Mary calls?

Exercise 9: Classification Errors

Given the following confusion matrix (system A):

	actual +	actual -
predicted +	100 True Positives	20 False Positive
predicted -	10 False Negatives	30 True Negatives

What is the Precision and Recall for this classification?

Precision = TP/TP+FP = 100 / 120

Recall = TP/TP+FN = 100/110

Exercise 10: Text feature preparation

What will the following sentences become after removing punctuation, stemming and stopwords?

a) Lisa is a student working on Artificial Intelligence.
b) Shuai was feeling lonely.
c) The intelligent systems course had a TA and many students.

Something like:

a. Lisa student work artificial intelligen
b. Shuai feel lone
c. Intelligen system course TA many student

Exercise 11: k-NN classification

Sentence a and c from the previous Exercise 10 are from the Canvas. Sentence b is from a diary. Let us use these examples as our training sets for a classifier that predicts whether a sentence is from Canvas or a diary. Considering the following sentence:

There are 200 students enrolled in the Intelligent Systems course.

Using K-NN classification with hamming distance, what are the two nearest neighbours for this sentence? And thus, what shall be the label if you need to predict the source of this sentence?

If you use hamming distance you have to count the number of non-agreeing feature values, so turn the sentences into features first, e.g. (alphabetic is also an option, but not necessary):

< Lisa,student,work,artificial,intelligen,Shuai,feel,lone,system,course,TA,many, 200, enrol>

And vectors

a) <1,1,1,1,1,0,0,0,0,0,0,0,0,0>

b) <0,0,0,0,0,1,1,1,0,0,0,0,0,0>

c) <0,1,0,0,1,0,0,0,1,1,1,0,0,0>

Now, the new sentence is transformed into a feature vector:

d) <0,1,0,0,1,0,0,0,1,1,0,0,1,1>

And just simple count the distinct values and normalise by the number of values 13 to see that a) and c) are the closest neighbours, so their class is taken over, and the prediction is that this new document is from Canvas.